CSE4096 Introduction to Deep Learning

Course Information

CSE4096 Intro. to Deep Learning

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Prerequisites

- Calculus
- Probability and statistics
- Linear Algebra
- Programming (Python)
- Basics of Machine Learning

Course Objectives

- Undergraduate level deep course in machine learning
- Give an overview of basic concepts in deep learning:
  - Multi-layer perceptrons
  - Convolutional Neural Networks
  - Recurrent Neural Networks
  - Deep Generative Models
- The students who successfully complete the course will be able to apply these concepts to real-world problems.

Learning Outcomes

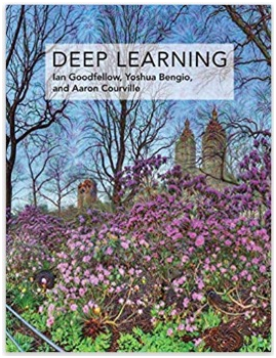
- **LO1:** Explain theoretical mathematical concepts of deep learning (optimization, regularization etc.)
- **LO2:** Describe and apply multi-layer perceptrons to machine learning problems that involve classification and regression.
- **LO3:** Explain basic concepts of convolutional neural networks (convolutions, padding, stride, pooling etc.).
- **LO4:** Explain modern CNN architectures (AlexNet, VGG GoogLeNet, ResNet etc.)

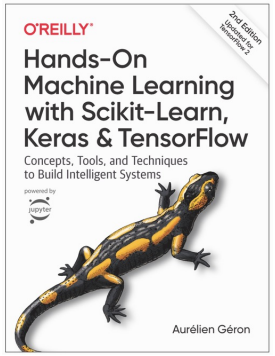
Learning Outcomes

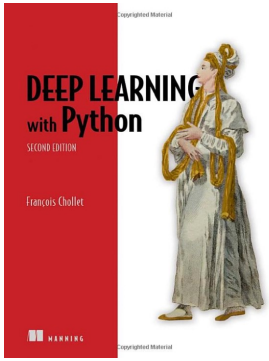
- **LO5:** Describe and apply CNNs to machine learning problems in computer vision (e.g. classification, object detection, semantic segmentation, face recognition).
- **LO6:** Describe and apply recurrent neural networks to machine learning problems that involve time series problems.
- **LO7**: Describe the basics of generative deep models.

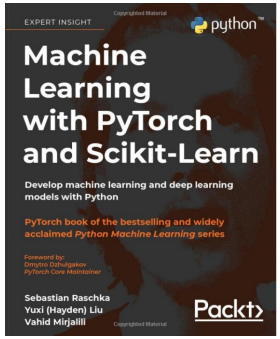
Reference Books

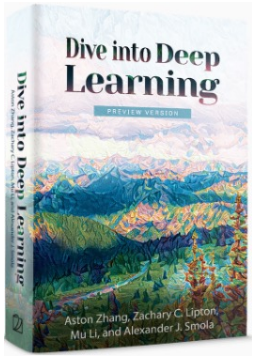
- **Deep Learning**, Ian GoodFellow, Yoshua Bengio, Aaron Courville, MIT Press, 2016. (available online: https://www.deeplearningbook.org/)
- **Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow**: Concepts, Tools, and Techniques to Build Intelligent Systems, Aurelien Geron, O'Reilly, second edition, 2019. ISBN 978-1-492-03264-9
- **Deep Learning with Python,** Francois Chollet, Second Edition, Manning Publishing, 2021, ISBN: 9781617296864
- **Machine Learning with PyTorch and Sckit-Learn,** Sebastian Raschka, Yuxi (Hayden) Liu, Vahid Mirjalili, Packt Publishing, 2022, ISBN 978-1-80181-931-2.
- **Dive into Deep Learning (online book),** https://d2l.ai/
- **Understanding Deep Learning, Simon J. D. Prince,** to be published by MIT Press. https://udlbook.github.io/udlbook/



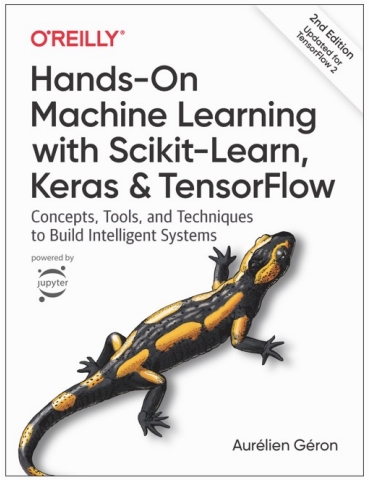






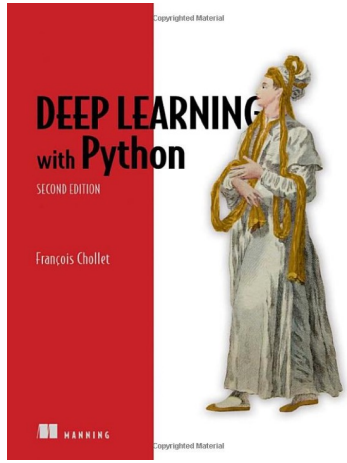


Reference Book



- **Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow,** Aurelien Geron, O'Reilly, second edition, 2019.
  - There is a series of Jupyter notebooks: code examples and exercises, available for download at https://github.com/ageron/handson-ml2
  - Web page for this book: errata, examples, and any additional information. https://homl.info/oreilly2
  - You can use Colab, but to install Jupyter, NumPy, pandas, Matplotlib and Scikit-Learn to your computer follow the steps given on pages 75 – 79 of the book.

Reference Book



- **Deep Learning with Python,** Francois Chollet, Second Edition,
- Manning Publishing, 2021.
  - Author is the creator of Keras
- All code examples in this book are available from the Manning website (which also contains errata):
  - https:// www.manning.com/books/deep-learning-with-python-second-edition

  OR on Github:
  - https://github.com/fchollet/deep-learning-with-pythonnotebooks
- They can be run directly in your browser via Google Colaboratory, a hosted Jupyter notebook environment that you can use for free.
- An internet connection and a desktop web browser are all you need to get started with deep learning.
- liveBook discussion forum: make comments, ask technical questions etc.
  - https://livebook.manning.com/#!/book/deep-learningwith- python-second-edition/discussion

Introduction to Deep Learning

- **Grading:**
  - Quizzes/Polls % 10
  - Midterm exam % 35
  - Project % 15
  - Final Exam % 40

Homeworks

- Homeworks will be given for your self-study, but they will not be graded.

- However, the students will be responsible from the concepts covered in the homeworks.

Projects

- Topic: Any topic related to the course.
- Phases:
  - Topic selection and proposal
  - Midterm report
  - Final report
  - Final presentation and demo
- Start thinking about your topic!

Web Site

- The class web page is available at: https://classroom.google.com
- Class code: ebholth
  - Use a gmail address showing your name and surname
  - Please click on the plus sign at top right corner to join the class using the class code
  - To access the lecture notes and homeworks click on the “Classwork” tab at the top of the page.

Attendance

- Attendance to at least 70% of the lectures in mandatory to pass the course.
- Otherwise your letter grade from the course will be DZ.

Academic Honesty

- Violations of scholastic honesty include, but are not limited to cheating, plagiarizing, fabricating information or citations, facilitating acts of dishonesty by others, having unauthorized possession of examinations, submitting work of another person or work previously used without informing the instructor, or tampering with the academic work of other students.

- In case academic dishonesty is observed, the first authority is the instructor of the course. The instructor may decide to give the student zero for the homework(s)/lab(s)/exam(s), give the letter grade FF, or may take disciplinary action.

Other Similar Courses

- Coursera (Andrew Ng)
  - Deep Learning Specialization
- University of Michigan (Justin Johnson)
  - Deep Learning for Computer Vision
- METU (Emre Akbaş)
  - Deep Learning
- New York University (Yann LeCun)
  - Deep Learning
- MIT (Alexander Amini, Ava Amini, Sadhana Lolla)
  - Introduction to Deep Learning